# Master Document 2: Team Matrix Organization & Work Distribution

**Project Name:** Agri-Grid: Smart Agriculture & Food Security Ecosystem
**Project Lead:** Ameer Muawiyah (Roll No: 1512)
**Context:** 7-Person Software Engineering Team

## 1. The "Matrix Approach" & Organizational Strategy

In a university setting, development teams frequently fall into the "solo full-stack" trap, where each member is assigned one module to build entirely on their own (database, backend, and frontend). For a highly interconnected system like Agri-Grid, this approach is a dangerous anti-pattern that leads to a "Frankenstein" application plagued by database conflicts, duplicated code, and disjointed user interfaces.

To prevent technical debt and ensure a cohesive final product, the Agri-Grid team employs the **Matrix Approach**, operating under a **"Two Hats"** philosophy. Every team member acts in two distinct capacities:

- **Hat 1: The Module Owner (Domain Expert):** Each of the 7 members is assigned exactly one of the 7 system modules. They are responsible for the initial domain research, brainstorming, and writing the rough drafts of the requirements, use-cases, and features for their specific module. This ensures every student has a clear, gradable contribution to the system's design.
- **Hat 2: The Global Role (Chief Editor & Executor):** When the project moves from brainstorming to formal documentation and coding, members switch to their specialized technical strengths. They act as the "Chief Editor" for a specific phase of the software engineering lifecycle across the *entire* application.

## 2. Domain Ownership: The 7 System Modules

To satisfy the "Hat 1" requirement, the system is segmented into 7 distinct domains. Each team member will select one of the following to research and draft:

1. **The Farmer’s Dashboard (Supply Side):** The CRUD-based digital ledger for farmers to log active crops, planting dates, and acreage, serving as the system's primary data generation point.
2. **The Predictive Yield Engine (Data & Analytics):** The mathematical core that pulls real-time environmental data to calculate harvest readiness and yield volume.
3. **The B2B Direct Marketplace (E-Commerce):** The digital storefront featuring a searchable catalog, escrow checkout logic, and digital contracts for wholesale buyers.
4. **The Cold-Chain Logistics Tracker (Transporter):** The mobile-first interface for dispatch manifests, QR custody handshakes, and live IoT temperature telemetry logging.
5. **The Agronomist Tele-Consult (Telemedicine):** The remote support portal where farmers submit diagnostic tickets and experts review imagery alongside API weather data to issue digital prescriptions.
6. **The Compliance & Subsidy Ledger (Admin):** The highly secure auditing portal used to

cross-check sustainability claims against API data and manage financial grant disbursements.

7. **The Consumer Transparency Portal (Public Verification):** The unauthenticated, mobile-friendly QR-scanning interface that generates visual farm-to-table provenance storyboards for grocery store consumers.

# 3. Global Execution: The 8 Engineering Roles

To satisfy the "Hat 2" requirement and build a cohesive system, the technical execution is divided into 8 specialized roles. Because the team consists of 7 members, the Project Lead takes on two complementary roles to balance the workload and control both the timeline and the core data architecture.

## Role 1: Project Manager (Assigned to: Ameer Muawiyah)

- **Planning Phase:** Sets the Agile sprint deadlines, manages the Jira/Trello boards, and oversees the overall project timeline to ensure all deliverables are met.
- **Coding Phase:** Tracks daily development progress, unblocks team members who are stuck, and ensures the 7 different modules merge together on schedule.

## Role 2: Lead Data Architect (Assigned to: Ameer Muawiyah)

- **Planning Phase:** Researches external data sources (Weather/Soil APIs) and designs the mathematical logic for the system's predictive features.
- **Coding Phase:** Bypasses frontend web design to focus purely on backend logic. Writes the Python data pipelines, extracts/cleans API data, and builds the algorithmic core (e.g., the predictive yield forecaster and environmental verification engine).

## Role 3: Requirements Engineer & Scrum Master

- **Planning Phase:** Acts as the "Chief of Requirements." Collects all 7 module rough drafts, resolves contradictions, standardizes formatting, and authors the formal Software Requirements Specification (SRS) document, defining all functional and non-functional constraints.
- **Coding Phase:** Acts as the Scrum Master. Continuously verifies that every feature the developers code matches the strict rules written in the SRS, actively preventing "scope creep."

## Role 4: Lead Software Architect & API Designer

- **Planning Phase:** Acts as the "Chief of Blueprints." Uses UML tools (StarUML/Draw.io) to redraw all 7 members' rough diagrams into master system blueprints (Use Case, Sequence, and Class diagrams), ensuring data flows logically between roles.
- **Coding Phase:** Designs the internal API structure. Sets the strict rules and endpoints for how the frontend web pages will request and receive data from the backend server.

## Role 5: Database Administrator (DBA)

- **Planning Phase:** Acts as the "Chief of Data Storage." Merges the team's data needs into master Entity-Relationship Diagrams (ERDs), mapping out how tables (Farmers, Crops, Escrow, Deliveries) connect without duplication.
- **Coding Phase:** Writes the complex SQL queries to create the unified database schemas. Ensures the database is normalized (ACID compliant) and can handle heavy analytical

JOIN queries.

## Role 6: Core Backend Engineer

- **Planning Phase:** Maps out the system's security architecture, focusing on Role-Based Access Control (RBAC) token routing and the multi-factor authentication flow.
- **Coding Phase:** Builds the actual web server (using a framework like Flask, Django, or Node.js). Constructs the secure routing engine that connects the DBA's database to the frontend, isolating the Data Architect's math from the public web.

## Role 7: UI/UX & Frontend Developer

- **Planning Phase:** Acts as the "Chief of Design." Creates the visual wireframes and dashboard prototypes for all 6 user roles, ensuring the entire ecosystem shares a single, cohesive, modern theme.
- **Coding Phase:** Writes the HTML, CSS, and JavaScript. Responsible for creating intuitive, mobile-responsive interfaces (crucial for the Transporter and Farmer roles) based on data provided by the Backend Engineer.

## Role 8: QA Lead & DevOps Engineer

- **Planning Phase:** Sets up the centralized GitHub repository, enforces branch protection rules (preventing direct pushes to main), and writes formal test cases based on the SRS document.
- **Coding Phase:** Takes charge of Verification and Validation (V-Model). Writes automated testing scripts (JUnit/Selenium) and manually stress-tests the application to locate vulnerabilities and bugs before final academic submission.